

Plan the bed. Walk the farm.

FarmFriend is a **voxel digital twin for real gardens and small farms**. Draw a scale-true blueprint of your plot, watch it stand up as a living 3D world, rehearse the season before you plant — then head outside and grow the real thing.

[BLUEPRINT 2D](#)[WORLD 3D](#)[SEASON SIMULATIONS](#)[LIVE WEATHER](#)[PLANTING CALENDAR](#)

YOUR PLAN, STANDING UP — A DRAWING IN THE BLUEPRINT BECOMES A WALKABLE FARM IN WORLD VIEW. EVERY BED, PLANT, POND AND BARN COMES FROM YOU.

Welcome to FarmFriend

Most garden planning happens in a spreadsheet — or in your head. FarmFriend gives your garden a **twin**: a living model of your beds, crops and season that you can walk through and experiment on before a single seed goes in the ground.

STEP 1 — DRAW IT

The Blueprint

A scale-true map of your plot where every cell is 25 cm. Spacing mistakes happen on screen — not in soil.

STEP 2 — WALK IT

The 3D World

One toggle and your plan stands up as a voxel farm — beds, greenhouses, ponds, animals, sky and weather.

STEP 3 — REHEARSE IT

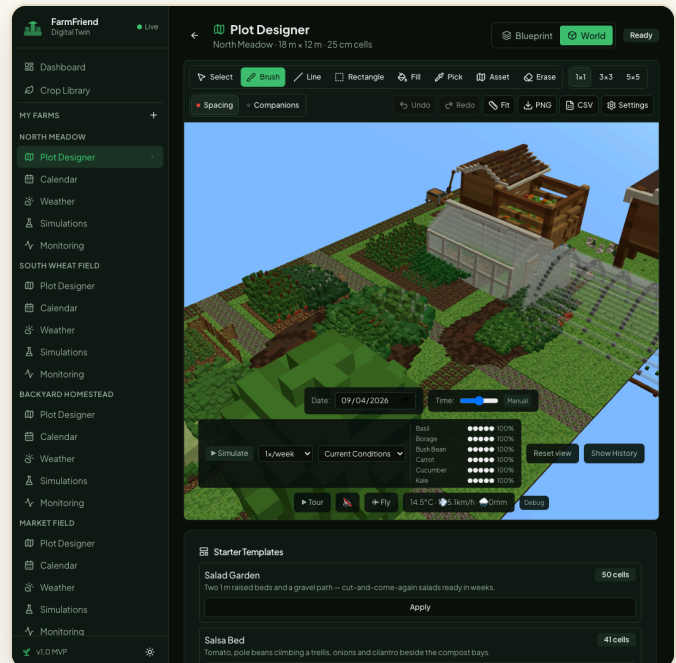
Season simulations

Run the season before it happens. Drought year? Heatwave? Extra fertilizer? Compare, then choose.

STEP 4 — GROW IT

The real thing

Print the plan, export the seed list, follow the frost-safe calendar — and plant with confidence.



SAME PLAN, STANDING UP — BEDS, GREENHOUSE, POND AND CROPS, STRAIGHT FROM THE DRAWING.

Good to know

Your plans save automatically in your browser — no account, no cloud, no setup. FarmFriend works offline and your data stays on your device. It's free and open source, and it runs on anything with a modern browser — laptop or tablet.

Start here: create your farm

A farm is your home base — it holds the plan, the weather station and the calendar. You can keep as many as you like: a backyard, an allotment, a field.

1 Add your farm

From the Dashboard, click **Add Farm** and give it a name you'll recognise.

2 Find your spot on the map

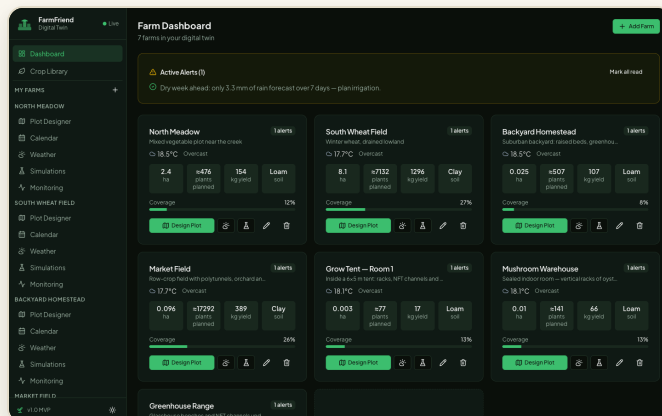
Type your town or village — FarmFriend fills in the location, which drives your live weather.

3 Check your frost dates

Last and first frost are estimated from your latitude and elevation. They shape your whole planting calendar — adjust them if you know your microclimate better.

4 Set the plot size

From a 4×3 m backyard bed up to a 60×60 m field. You can change it any time in Settings.



ADD FARM — TOP RIGHT OF THE DASHBOARD. EACH FARM CARD ALREADY ESTIMATES PLANTS, YIELD AND SOIL FROM YOUR PLAN.

Tip

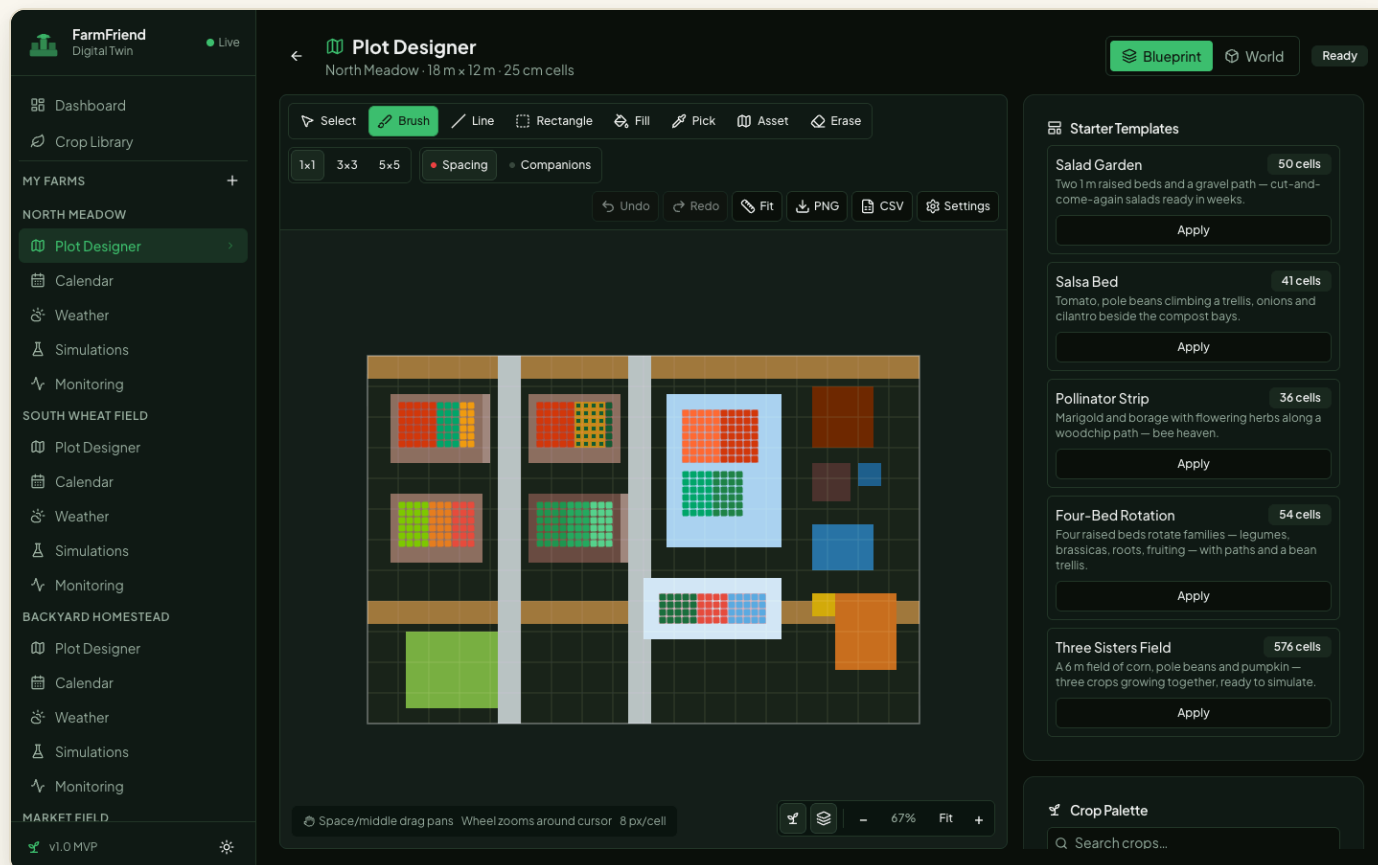
Not sure where to start? Apply a **starter template** in the Plot Designer — Salad Garden, Salsa Bed, Pollinator Strip, Four-Bed Rotation or Three Sisters Field — then rearrange freely. One click of Undo restores the previous plan.

Good to know

FarmFriend ships with a handful of **demo farms** — a market field, a greenhouse range, even a mushroom warehouse. Poke around in them as much as you like; when you add your own farm it keeps its own plan, weather and calendar.

Draw your plan in the Blueprint

The Plot Designer is where your farm takes shape: a canvas of 25 cm cells with real crops, real structures and real spacing math — so what you draw is what you can plant.



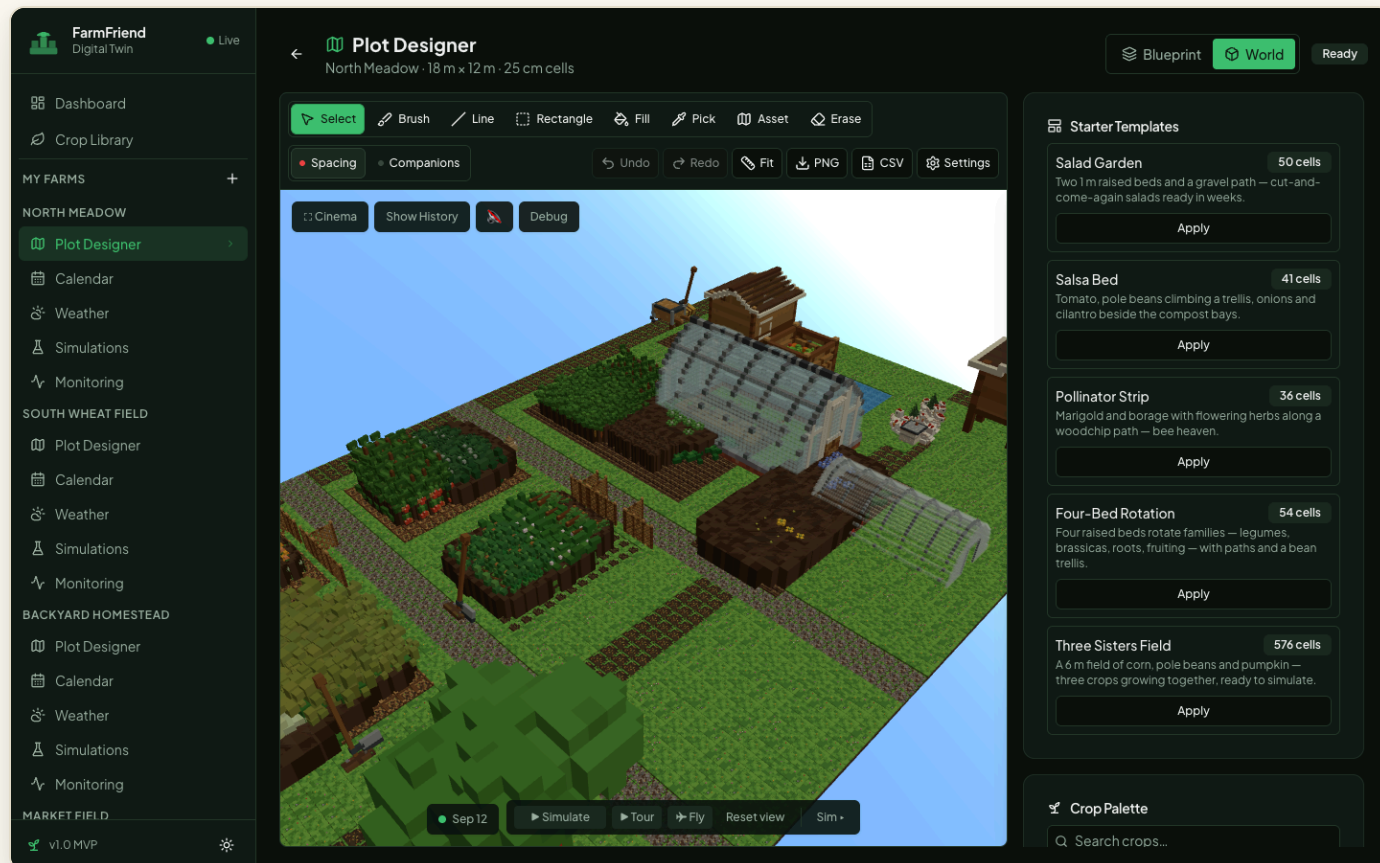
THE PLOT DESIGNER — TOOLS UP TOP, STARTER TEMPLATES AND PALETTES ON THE RIGHT. SPACING AND COMPANION CHECKS RUN LIVE WHILE YOU DRAW.

- **Paint the ground first.** Raised beds, paths, sheds, greenhouses, fences, ponds and water taps — every structure has its true footprint.
- **Plant on top.** Pick crops from the palette and paint them in — brush sizes 1×1, 3×3 and 5×5, plus Line and Rectangle for quick beds.
- **Let it catch mistakes.** Crowded plants are flagged for spacing; crops that dislike each other are outlined; good neighbours get a green halo.
- **Watch the numbers.** Plant counts, estimated yield and daily water demand update as you draw.
- **Scrub with confidence.** Layer toggles hide plants or ground, zoom controls keep you oriented, and Undo/Redo remembers every stroke.
- **Take it with you.** Export a print-ready PNG of the plan, or a CSV seed-shopping list with quantities and sowing windows.

Shortcuts **V** select **B** brush **R** rectangle **A** asset **E** erase **I** pick **G** fill **L** line **⌘Z** undo

Walk your farm in 3D

Flip from Blueprint to **World** and your drawing stands up. Orbit around it, zoom in to a single bed, and watch the season move. It's the same plan — just alive.



WORLD VIEW — THE NORTH MEADOW PLAN STANDING UP AT MIDDAY. THE DATE CHIP AND PLAYBACK CONTROLS LIVE ALONG THE BOTTOM EDGE; ORBIT, ZOOM AND FLY WITH THE MOUSE.

- **Scrub the date** and every crop grows through six stages — seedling to harvest. Plan the harvest, not just the planting.
- **Drag the time of day.** Dawn, noon, golden hour, night — the sun, shadows and sky follow the real clock and weather.
- **Paint a barn, get livestock.** Cows, pigs and sheep arrive around a barn; hens scratch by the coop; bees and butterflies visit flowers in bloom.
- **Take the guided tour** for a smooth cinematic fly-over of your plan — great for showing off.
- **Or fly it yourself.** Flight mode lets you soar over your own farm like a hawk.
- **Live weather, indoors and out.** If it's overcast at your farm, the world is overcast too. Rain falls. Plants sway.

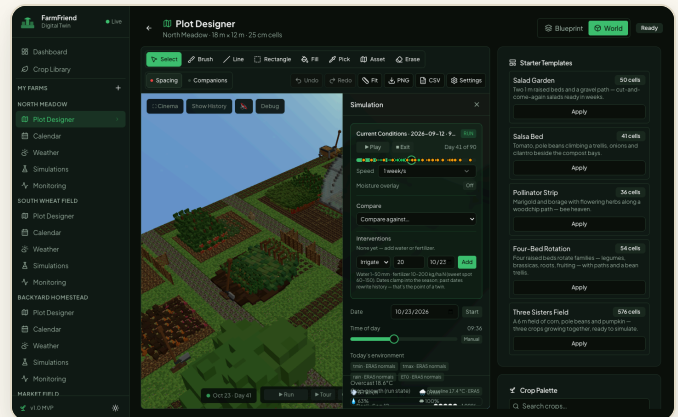
Tip

Hit **Cinema** to hide every panel for a full-screen walkthrough — and Show History to see ghost echoes of what you've changed. You can even plant and erase right in the 3D world; your edits flow straight back into the Blueprint.

Rehearse the season before it happens

February is for simulations. A **run** plays your farm through a whole season — day by day, using your plan, your weather and your soil — so you can test ideas while they're still cheap.

- 1 Open Simulations → New Run**
Pick a scenario: **Current Conditions**, **Drought Stress**, **Heat Stress** or **Optimal Input**.
- 2 Fine-tune the weather**
Slide temperature (−10 to +10 °C) and rainfall (0–300 %) to build exactly the year you're worried about — or curious about.
- 3 Create the run**
Choose a start date and a length of 60–365 days. Each run is recorded so you can come back to it.
- 4 Press Play in the world**
Watch the season unfold at a week per second — crops grow, weather rolls in, and struggling plants visibly yellow under stress.



MID-SEASON, DAY 41 — THE DRAWER TRACKS THE DAY, WEATHER AND STRESS. ADD IRRIGATION OR FERTILIZER AS INTERVENTIONS AND WATCH WHAT CHANGES.

- Interventions rewrite history — on purpose.** Add water or fertilizer on any date, past or future, and the run refolds to show the difference. That's the point of a twin.
- See the soil.** The moisture overlay shows ground drying out; the timeline marks stress and pest pressure day by day.

DAY TIMELINE

PLAY & SPEED

MOISTURE OVERLAY

INTERVENTIONS

COMPARE AGAINST...

Compare the seasons, pick your plan

Every run keeps its scorecard — yield, rain, irrigation, plant stress. Compare them and the better choice usually picks itself.

Simulation Runs
North Meadow — deterministic runs over this farm's plan, weather and soil

Comparing 2 runs — deltas vs Drought Stress - 2026-09-12 - 90d - s5cw
Primary = first selected (Drought Stress - 2026-09-12 - 90d - s5cw), ghost = second (Current Conditions - 2026-09-12 - 90d - s4fr).

Metric	BASE Drought Stress - 20...	Open in world	Current Conditions - 2026...	Open in world
Yield kg	643.27	601.13	-42.14	
Water use mm	114.6	116.6	+2.0	
Stress cell-days Σ of the four terms	8605	13453	+4848	
Outbreak cell-days pest ≥ 0.6	0	0		
Residual nitrogen kg/ha N	427.8	135.0	-292.8	

Deltas are colored by direction, not judgment: green = better (more yield, less water, stress or outbreak), red = worse, gray = neutral (residual nitrogen has no inherent good direction). Stress cell-days = one cell under stress for one day, summed over the four stress terms.

Run Comparison

Run	Scenario	Days	Yield (kg)	Rain (mm)	Irrigated (mm)	Stress (cell-days)
Drought Stress - 2026-09-12 - 90d - s5cw	drought	90	643.27	261.5	0	8605
Current Conditions - 2026-09-12 - 90d - s4fr	baseline	90	601.13	523	0	13453

Drought Stress - 2026-09-12 → 2026-12-10 (drought) [complete]

Yield	643.27 kg	Rain	261.5 mm
Irrigated	0 mm	ETO	143.5 mm
Stress	8605 cell-days	Mature	414 / 414 cells

Current Conditions - 2026-09-12 → 2026-12-10 (baseline) [complete]

Yield	601.13 kg	Rain	523 mm
Irrigated	0 mm	ETO	143.5 mm
Stress	13453 cell-days	Mature	383 / 414 cells

TWO 90-DAY RUNS, SIDE BY SIDE — DELTAS ARE COLOUR-CODED: GREEN MEANS BETTER (MORE YIELD, LESS WATER OR STRESS), RED MEANS WORSE. OPEN BOTH IN THE WORLD TO WATCH THEM PLAY TOGETHER.

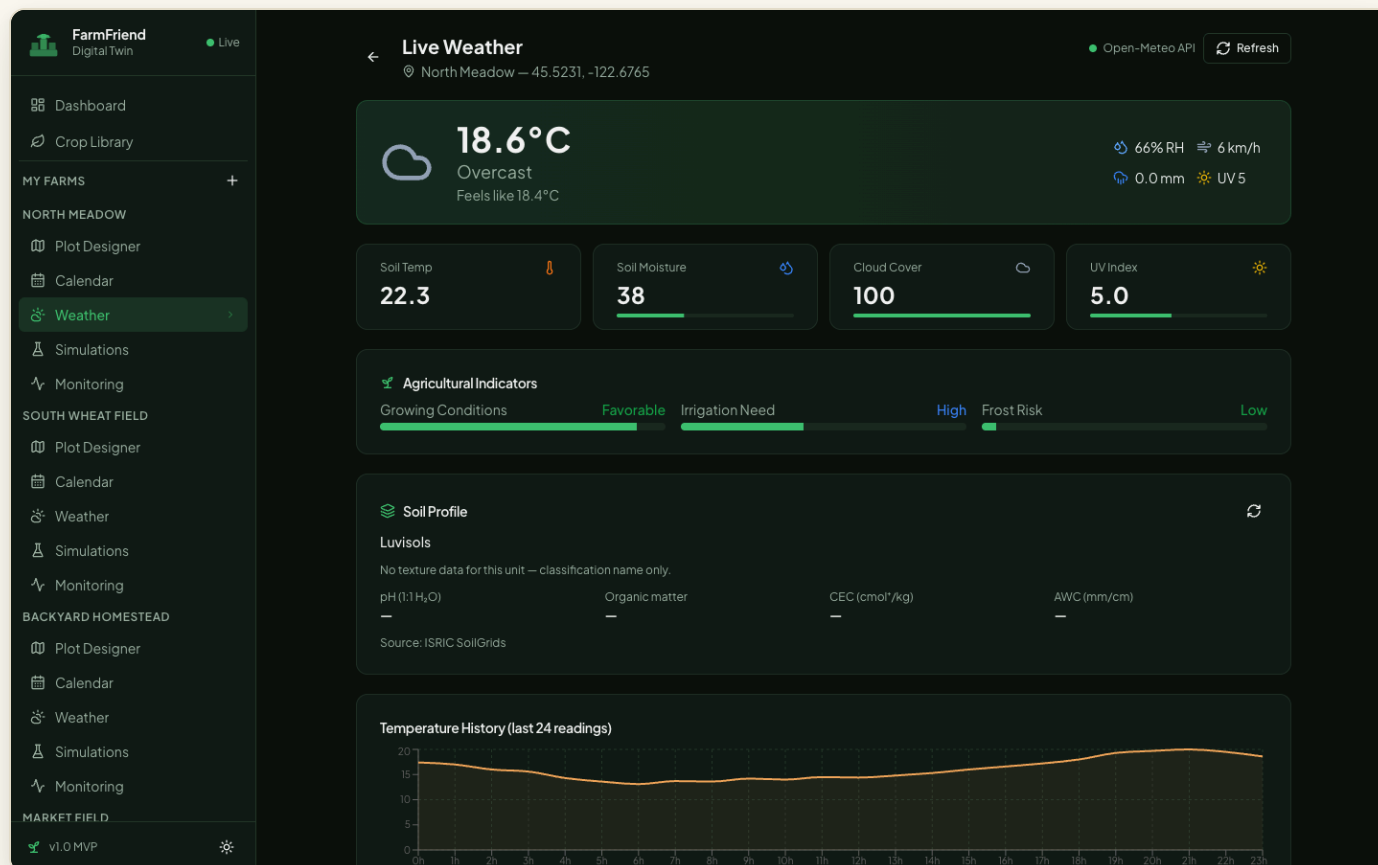
- **Tick any two runs** — up to six — and the comparison table shows exactly what changed and by how much.
- **A/B them in the world.** A translucent ghost of the second run grows alongside yours, so you can see the difference, not just read it.
- **Every number is explained.** Click into a run for a day-by-day breakdown of water, heat, cold, nitrogen and pest pressure — per bed.
- **Same inputs, same result.** Runs replay deterministically, so an experiment from last month still means what it meant then.

A twin, not a crystal ball

Simulations are careful what-if estimates built on real crop data and historic weather patterns — brilliant for **comparing plans**, not a guarantee of harvest. Use them to choose between options, then let the season have the final word.

Read the sky before you sow

Each farm gets its own live weather station — real conditions and a 7-day forecast for your exact location, translated into terms a gardener can act on.



LIVE WEATHER — CURRENT CONDITIONS, SOIL TEMPERATURE AND MOISTURE, AGRICULTURAL INDICATORS AND THE SOIL PROFILE FOR YOUR SPOT ON THE MAP.

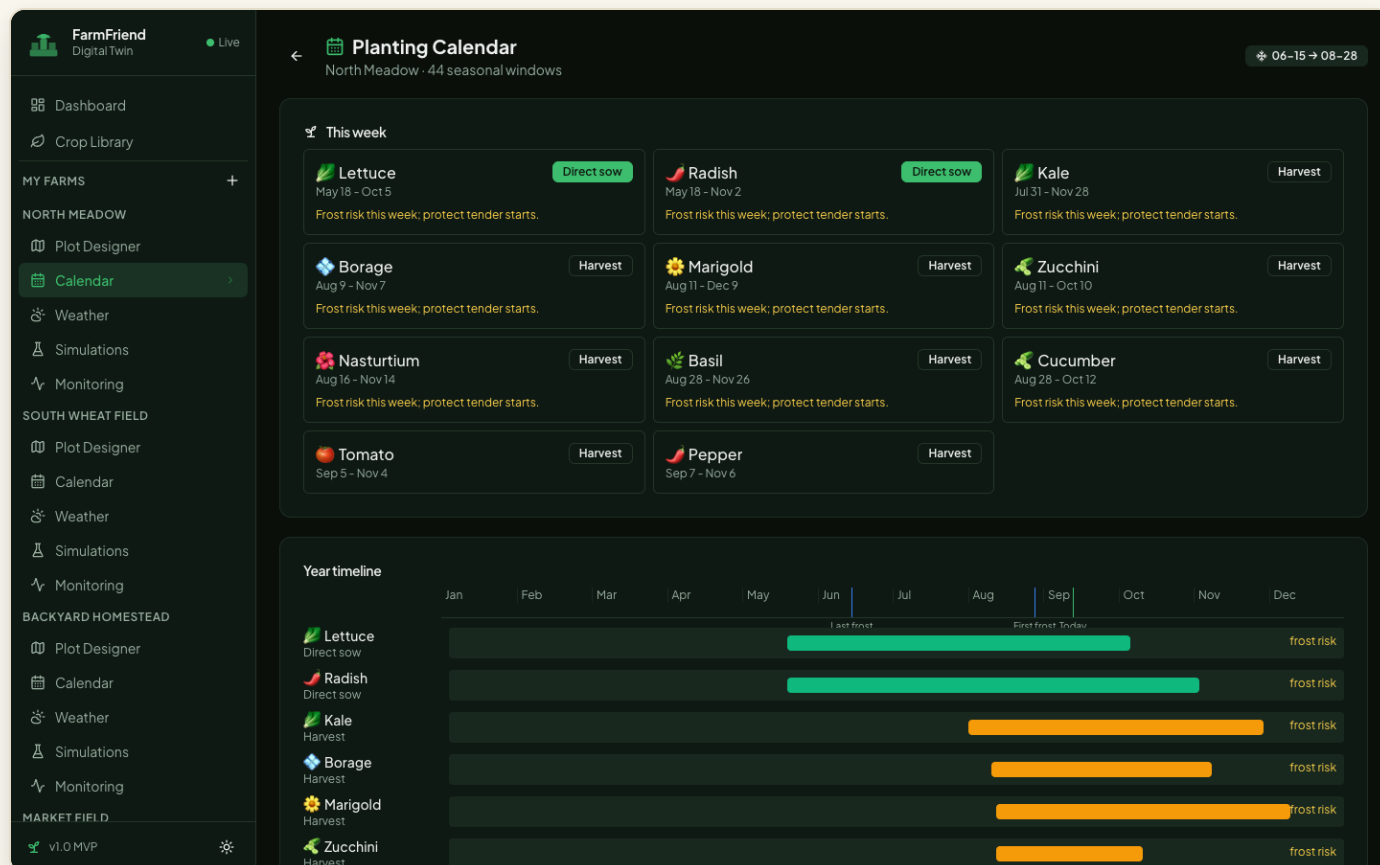
- **Soil first.** Soil temperature and moisture at surface level — the two numbers that tell you whether it's actually time to sow.
- **Alerts before they bite.** Frost, heatwaves, heavy rain and dry spells are flagged from the real forecast, per farm.
- **Know your ground.** The soil profile shows the classification and qualities of the earth under your farm.
- **Offline-friendly.** No internet? The world keeps going on the last known weather, clearly marked as cached.

Tip

The Dashboard collects alerts for **all** your farms in one banner — a dry week ahead, a cold snap coming — so one glance is enough before you head out.

Stay ahead of the frost

Your frost dates plus each crop's real needs become a year-round schedule: sow indoors, transplant out, direct sow and harvest — computed for every crop in your plan.



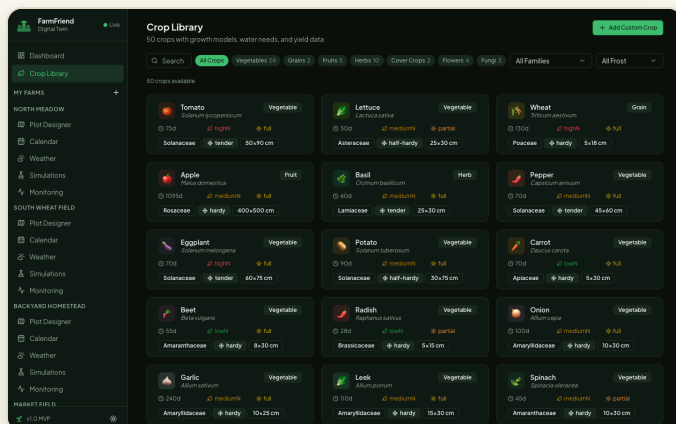
THE PLANTING CALENDAR — “THIS WEEK” TELLS YOU WHAT TO DO RIGHT NOW; THE YEAR TIMELINE LAYS OUT EVERY CROP’S WINDOWS BETWEEN LAST AND FIRST FROST.

- **“This week” is your to-do list.** What to sow, plant out and harvest right now — with frost-risk warnings on tender crops.
- **Built from your plan.** Change what you plant and the calendar updates itself — no double bookkeeping.
- **Every crop gets a bar.** Follow each variety across the year: indoor sowing, transplant, direct-sow window, harvest season.
- **Frost markers included.** Last and first frost dates are drawn on the timeline so risk is visible at a glance.

Tip

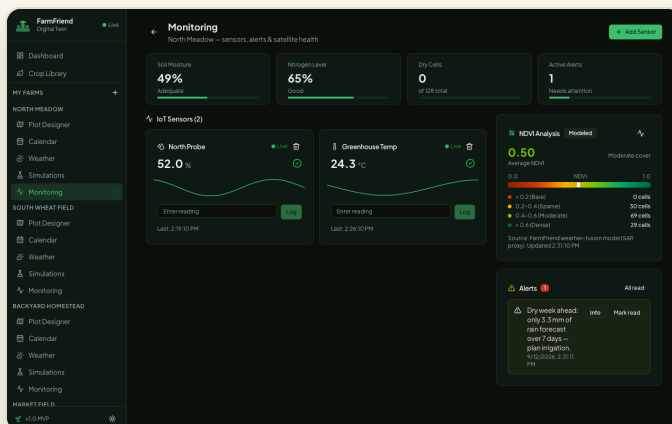
Frost dates are estimated from your location — if your garden runs warm or cold, override them in your farm settings and every window shifts to match.

Everything else you'll reach for



CROP LIBRARY — 50 CROPS WITH THE NUMBERS THAT MATTER.

- **A crop library with real data.** 50 crops with spacing, frost tolerance, companion pairings and yield estimates — searchable and filtered by family.
- **Monitoring.** A per-farm health view that gathers condition readouts in one place.
- **Yours to keep.** Export the plan as PNG, the seed list as CSV — and your work stays saved in your browser automatically.



MONITORING — A HEALTH SNAPSHOT FOR EVERY FARM.

Questions people ask

Where does my data live? In your browser, on this device. Nothing is uploaded.

Does it cost anything? No — FarmFriend is free and open source (MIT).

Tablet? Yes — draw with touch, pinch to zoom.

Is it a game? It looks like one. The numbers underneath are real horticulture.

Plan the bed. Walk the farm.

HAPPY GROWING — FROM THE FARMFRIEND TEAM